

Raffles Institution
Year 5-6 Physics Department

2024 Year 6 H2 Physics Term 3 Timed Practice Solution

Section A

Qn	Ans	Solution
1	A	$g_s = \frac{GM_p}{R^2}$ $g = \frac{GM_p}{(R+2R)^2} = \frac{1}{3^2} \frac{GM_p}{R^2} = \frac{1}{9} g_s$
2	B	Gravitational force on the satellite provides the centripetal force. $\frac{GM_p m_s}{r^2} = m_s \frac{v^2}{r}$ kinetic energy of satellite, $E_k = \frac{1}{2} m_s v^2 = \frac{1}{2} \frac{GM_p m_s}{r}$ gravitational potential energy of satellite, $E_p = -\frac{GM_p m_s}{r}$ 1st satellite: $E_{p1} = -\frac{GM_p m_s}{r} = -3.2 \text{ MJ}$ 2nd satellite: $E_{p1} + E_{k1} = -\frac{GM_p m_s}{2r} + \frac{1}{2} \frac{GM_p m_s}{2r}$ $= -\frac{1}{2} \frac{GM_p m_s}{2r}$ $= -\frac{1}{4} \frac{GM_p m_s}{r}$ $= \frac{1}{4} (-3.2 \text{ MJ})$ $= -0.80 \text{ MJ}$
3	A	Gravitational force on each star provides the centripetal force. net force = gravitational force = $\frac{GMM}{(2R)^2} = \frac{GM^2}{4R^2}$

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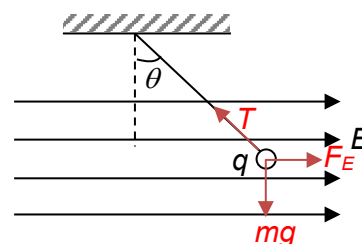
4	C	Since $a = -\omega^2 x$, gradient of a-x graph $= -\omega^2$. $v = \pm \omega \sqrt{x_0^2 - x^2}$ speed at $x = 0.10$ m, $ v = \omega \sqrt{x_0^2 - x^2}$ $= \left(\sqrt{\frac{3.0}{0.30}} \right) \left(\sqrt{0.30^2 - 0.10^2} \right)$ $= 0.89443 = 0.89 \text{ m s}^{-1}$
5	A	Take direction downwards from equilibrium as positive. When $t = 0$, since block is at lowest point, its displacement from equilibrium is $x = x_0 \cos \omega t$. resultant force on the block, $mg - N = ma$ $N = mg - ma = mg - m(-\omega^2 x) = mg + m\omega^2 x_0 \cos \omega t$ Hence, N varies with t as a cosine function, translated upwards by mg. N is always greater than zero, as the block is always in contact with the board.
6	D	The period T of oscillation of a pendulum of length L is given by $T = 2\pi \sqrt{\frac{L}{g}}$ i.e. $T \propto \sqrt{L}$. As the sand trickles out of the bucket, the mass of the sand decreases and the effective length L of the pendulum increases as the centre of gravity of the bucket of sand lowers. Hence, the period will initially increase. However, the centre of gravity of the bucket of sand will eventually rise back up to its original position as the mass of the sand continues to decrease until the bucket becomes empty. This decreases the effective length L of the pendulum back to its original length. Hence, the period will decrease back to its original value.
7	A	$\lambda = \frac{v}{f} = \frac{340}{540} = 0.62963 \text{ m}$ $\frac{x}{\lambda} = \frac{\phi}{2\pi}$ $x = \frac{(3/4)\pi}{2\pi} \times 0.62963$ $= 0.23611 = 0.24 \text{ m}$
8	B	$I = ka^2 = \frac{P}{S} \Rightarrow P = ka^2 S$ $P' = k(4a)^2 \left(\frac{S}{4} \right)$ $= k(16a^2) \left(\frac{S}{4} \right)$ $= 4(ka^2 S)$ $= 4P$

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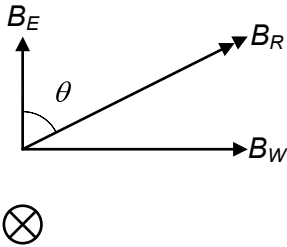
9	C	Another 6 pressure maximum signals detected means 6 internodal distances in 80 cm. distance between 2 displacement nodes is $\frac{\lambda}{2}$ $6\left(\frac{\lambda}{2}\right) = 80 \times 10^{-2}$ $\lambda = \frac{80 \times 10^{-2}}{3}$ $f = \frac{v}{\lambda} = \frac{330}{\frac{80 \times 10^{-2}}{3}} = 1237.5 = 1200 \text{ Hz}$
10	A	For first minimum of single-slit diffraction pattern, $b \sin \theta = \lambda$ Rayleigh criterion to just resolve the 2 lights, the central maximum of the diffraction pattern of one light coincides with the first minimum of the diffraction pattern of the other light. Hence minimum angular separation between the 2 objects $\theta_{\min} \approx \frac{\lambda}{b}$ The shortest wavelength of visible light will give the minimum value of d. $d_{\min} = r \theta_{\min} = r \left(\frac{\lambda_{\min}}{b} \right) = (400 \times 10^3) \left(\frac{400 \times 10^{-9}}{4.0 \times 10^{-3}} \right) = 40 \text{ m}$
11	B	$d \sin \theta = n \lambda$ $d = \frac{n(v/f)}{\sin \theta} = \frac{3}{\sin(82^\circ/2)} \left(\frac{3.00 \times 10^8}{550 \times 10^{12}} \right) = 2.4942 \times 10^{-6} \text{ m}$ 1 m ----- $\frac{1}{d}$ lines $1 \text{ mm} \text{ ----- } \frac{1}{d} \left(\frac{1}{10^3} \right) = \frac{1}{2.4942 \times 10^{-6}} \left(\frac{1}{10^3} \right) = 400.93 = 4.0 \times 10^2 \text{ lines}$
12	B	amount of thermal energy required, $E = mc\Delta T = 1.0 \times 4200 \times (50 - 20) = 126000 \text{ J}$ incident power on the mirror, $P = \text{intensity} \times \text{area} = 1400 \times 0.40 = 560 \text{ W}$ $P = \frac{E}{t}$ $t = \frac{E}{P}$ $= \frac{126000}{560}$ $= 225 \text{ s}$ $= \frac{225}{60} \text{ min} = 3.75 = 3.8 \text{ min}$

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13	D	Applying the first law of thermodynamics to this system, $\Delta U = Q + W = 0 + W = W$. Hence, work done on the gas increases the internal energy of the gas. Since for an ideal gas, its internal energy and kinetic energy is proportional to its temperature, the temperature of the gas increases and the kinetic energy of the molecules increases.
14	B	for a fixed mass of ideal gas, $pV = nRT \Rightarrow T \propto pV$ from X to Y: Since p is constant, $T \propto V$. Temperature increases linearly with volume. from Y to Z: Temperature decreases as $pV = 4000$ at Z is smaller than $pV = 6000$ at Y.
15	B	By definition, work done by external force is the electric potential energy of the system. $W_{F_{ext}} = U_{syst}$ $= U_{++} + 2U_{+-}$ $= \frac{1}{4\pi\epsilon_0} \left[\frac{e^2}{2a} + 2 \left(\frac{-e^2}{a} \right) \right]$ $= \frac{1}{4\pi\epsilon_0} \left(\frac{-3e^2}{2a} \right)$ $= \frac{-3e^2}{8\pi\epsilon_0 a}$
16	D	Since the ball is at rest, the electric force on the ball is in the direction of the electric field. This implies the charge on the ball is positive. Consider horizontal and vertical equilibrium. $\sum F_x = 0: T \sin \theta = qE$ ----- (1) $\sum F_y = 0: T \cos \theta = mg$ ----- (2) $\frac{(1)}{(2)}: \tan \theta = \frac{qE}{mg}$ $q = \frac{mg}{E} \tan \theta$
17	C	Since electric field lines point from a region of higher potential to a region of lower potential, the potential at Y is lower than the potential at X. Hence, as the positive charge moves from X to Y, its electric potential energy decreases. As the charge moves horizontally from Y to Z, it is moving along an equipotential line. Hence Y and Z are at the same potential. There is no change in potential energy as the charge moves from Y to Z. overall decrease in potential energy $= q(V_x - V_y)$ $= (3.0)(5.0 \times 3.0)$ $= 45.0 \text{ J}$



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18	C	Since P and Q have the same length x, the cross-sectional area of P and the total cross-sectional area of the wires that make Q must be the same as P and Q are made from pieces of constantan of the same volume. Using $R = \rho \frac{L}{A}$, for the same length and cross-sectional area, the resistance of P and Q must be the same. Mathematically, $R_P = \rho \frac{x}{A}$ R_Q consists of N number of resistors, each with resistance $\rho \frac{x}{A/N}$, connected in parallel. $R_Q = \left(\left(\rho \frac{x}{A/N} \right)^{-1} \times N \right)^{-1} = \left(\frac{A}{\rho x N} \times N \right)^{-1} = \rho \frac{x}{A}$ Hence, $R_P = R_Q$.
19	A	Since the filament of bulb X is broken, there is no current flowing through the branch with bulbs X and Y. There is no potential difference across bulb Y. The potential difference across bulb X is 10 V due to the break in the branch in bulb X, Bulb Z is in a separate parallel branch and current continues to flow through it. The potential difference across it will remain unchanged at 10 V. As the branch with bulbs X and Y no longer has a current flowing through it, the effective resistance of the circuit is due to the resistance of bulb Z alone. The effective resistance is now greater than the effective resistance when currents flow in both parallel branches. Thus, current through the main branch decreases.
20	C	 By right hand grip rule, magnetic field produced by wire at compass, B_W, points East. $B_W = \frac{\mu_0 I}{2\pi d} = \frac{4\pi \times 10^{-7} \times 3.0}{2\pi \times 9.0 \times 10^{-3}} = 6.6667 \times 10^{-5} \text{ T}$ $\theta = \tan^{-1} \frac{B_W}{B_E} = \tan^{-1} \frac{6.6667 \times 10^{-5}}{3.5 \times 10^{-5}} = 62.3 = 62^\circ$ 62° East of North.
21	C	Due to the interaction of the magnetic fields of the wires, the force is attractive between the wires. By Newton's 3rd law, $F_2 = F_1$ (action-reaction pair).

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22	C	Faraday's law: $E = -\frac{d\Phi}{dt}$ for the coil: $ E_{coil} = \left -\frac{d(NBA \cos \omega t)}{dt} \right = NBA\omega \sin \omega t$ peak e.m.f. in coil, $E_{coil,max} = NBA\omega = 2\pi NBAf$ for the metal disc: $ E_{disc} = \left -\frac{d(BA)}{dt} \right = B \frac{dA}{dt} = B \left(\frac{A}{T} \right) = BAf$ $\frac{ E_{coil,max} }{ E_{disc} } = \frac{2\pi NBAf}{BAf} = 2\pi N$
23	B	$I_{rms} = \sqrt{\frac{\int_0^T I^2 dt}{T}}$ $= \sqrt{\frac{(0.44)^2 (1.0) + (-0.67)^2 (0.5) + (-1.0)^2 (1.5)}{3.0}}$ $= 0.79959 = 0.80 \text{ A}$
24	A	$P = \frac{V_{rms}^2}{R} = \frac{V_0^2}{2R}$ Option A (correct): When R is increased to $4R$, $P' = \frac{V_0^2}{2(4R)} = \frac{1}{4}P$ Option B (incorrect): $P' = \frac{(V_0/4)^2}{2R} = \frac{1}{16}P$ Option C (incorrect): Frequency does not affect the root-mean-square values of the current and voltage. Hence frequency does not affect mean power dissipated. Option D (incorrect): By connecting a diode, the current through the resistor and the p.d. across the resistor will be half-wave rectified. Hence the mean power dissipated will be $\frac{1}{2}P$.
25	D	Option D (correct): Increasing the intensity, increases the rate of photons incident on the metal surface. This increases the rate of photoelectrons emitted from the metal surface, which increases the photocurrent. Option A (incorrect): Photoelectrons are emitted as long as the frequency of the incident radiation exceeds the threshold frequency of the metal surface, regardless of intensity. Option B (incorrect): Since $V_s = \frac{hf}{e} - \frac{\Phi}{e}$, doubling the frequency of the incident radiation, does not double the stopping potential. Option C (incorrect): Threshold wavelength is the maximum wavelength of the incident radiation below which emission of photoelectrons will occur.

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26	D	kinetic energy, $E = \frac{1}{2}mv^2 = \frac{1}{2} \frac{(mv)^2}{m} = \frac{1}{2} \frac{p^2}{m}$ momentum, $p = \sqrt{2mE}$ de Broglie wavelength, $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}}$ $\frac{\lambda_e}{\lambda_p} = \sqrt{\frac{2m_p E_p}{2m_e E_e}} = \sqrt{\frac{1.67 \times 10^{-27}}{9.11 \times 10^{-31} \times 5}} = 19.148 = 19$
27	A	The maximum loss in the kinetic energy of an electron in a single collision is converted to the energy of a photon of minimum wavelength. $eV = \frac{hc}{\lambda_{\min}}$ $\lambda_{\min} = \frac{hc}{e} \left(\frac{1}{V} \right)$ $\lg \lambda_{\min} = \lg \frac{hc}{e} - \lg V$ where $\lg \frac{hc}{e}$ is a constant The graph of $\lg \lambda_{\min}$ against $\lg V$ is a straight line with negative gradient.
28	D	no. of nucleons = P binding energy due to mass defect = $(\Delta m)c^2$ binding energy per nucleon = $\frac{c^2 \Delta m}{P}$
29	D	By the principle of conservation of linear momentum, initial momentum of ^{238}U = final momentum of ^{234}Th + final momentum of ^4He $0 = 234v_{\text{Th}} - 4v_{\alpha}$ $\frac{v_{\alpha}}{v_{\text{Th}}} = \frac{234}{4}$ $\frac{\frac{1}{2}m_{\alpha}v_{\alpha}^2}{\frac{1}{2}m_{\text{Th}}v_{\text{Th}}^2} = \frac{m_{\alpha}}{m_{\text{Th}}} \left(\frac{v_{\alpha}}{v_{\text{Th}}} \right)^2 = \left(\frac{4}{234} \right) \left(\frac{234}{4} \right)^2 = \frac{234}{4}$ $\frac{\frac{1}{2}m_{\alpha}v_{\alpha}^2}{E} = \frac{234}{238}$ kinetic energy of α-particle, $\frac{1}{2}m_{\alpha}v_{\alpha}^2 = \frac{234}{238}E = 0.98E$
30	C	true count rate = $602 - 18 = 584$ after 3 half-lives, true count rate = $\left(\frac{1}{2} \right)^3 \times 584 = 73$ count rate recorded by detector = $73 + 18 = 91$